

Topic 19: Modern Analytical Techniques II

In order to develop their practical skills, students should be encouraged to carry out a range of practical experiments related to this topic. Possible experiments include using chromatography to identify unknowns such as metals in a 'silver' coin or amino acids in a protein.

Mathematical skills that could be developed in this topic include calculating R_f values, interpreting a variety of different spectra, calculating relative molecular masses, using the $(n+1)$ rule for proton NMR.

Within this topic, students can consider a wider range of instrumental methods used for analysis, such as NMR; and see how this technique is used in medicine through MRI scans. They can also see a wide range of applications that rely on a combination of different analytical techniques.

Students should:
Topic 19A: Mass Spectrometry
1. be able to use data from mass spectra to: - i suggest possible structures of a simple organic compound given relative molecular masses, accurate to four decimal places - ii calculate the accurate relative molecular mass of a compound, given relative atomic masses to four decimal places, and therefore identify a compound